

► Moon Walk — *WHOOOP for your walking aid.*

WELLNESS, NOT MEDICINE

Live demo → moonwalkerscamper.netlify.app

Does not diagnose, treat, or predict.

A clip-on sensor that turns the cane or walker someone **already uses** into a gait instrument. Like a WHOOP for the way you walk, it does two things: it **tracks** how you normally walk and notices when that changes, and it **coaches** your rhythm in real time when you ask it to.

The problem

Millions walk with a cane or rollator because of a condition that changes over time — stroke recovery, Parkinson's, ageing, post-surgery rehab. **Nobody is watching the trend.** Subtle decline creeps in unnoticed; rehab progress goes uncelebrated; and by the next clinic visit nobody remembers how the walking changed. People have no objective, continuous picture of their own gait.

The solution — two pillars

- **Track — know your normal, notice the change.** Moon Walk clips onto the **Host Aid**, continuously measures **Gait** (cadence, rhythm, loading, symmetry, stride trends), learns that person's normal **Baseline on-device**, and watches for sustained **Drift**. When walking meaningfully changes it raises a gentle wellness **Alert** — *"your walking has changed; you may want to mention it to your doctor."* Awareness, not a diagnosis.
- **Coach — improve your rhythm in the moment.** Opt into **Training Mode** for real-time **Coaching Cues** — biofeedback by **voice, vibration, or LED** — guiding you toward *your own* target rhythm, like a running-cadence coach. Coaching, not correction: opt-in, never names a condition, *not* therapy.

The live demo — multi-user, real-time

Guests each walk a short course wearing a Nano 33 BLE. The IMU streams over **BLE → a local hub → the dashboard** (gait data stays local, never the cloud), which plots **every walker live and side by side**: a real-time signal trace, a live **cadence comparison**, and a leaderboard — then **Training Mode** coaches a chosen walker with voice/vibration/LED. Try it: moonwalkerscamper.netlify.app

Why it's different

The 2014 "Smart Walker" *replaced* the aid and streamed raw video to the cloud to navigate. Moon Walk **instruments the aid you own**, keeps the **intelligence and health data on-device**, and makes an **honest wellness claim** — no overclaim. Gait sensors on a walking aid track *change* reliably (temporal metrics ICC 0.72–0.97), so we lead with temporal numbers and show stride/velocity only as **relative trends**, never clinical absolutes.

Honest scope

- Not a medical device — no diagnosis, treatment, disease/fall-risk prediction; a clinician, never the device, interprets data.
- **Hardware in hand is IMU-only** (LSM9DS1): all six gait metrics derive from the IMU; loading/asymmetry (force sensor) and the **Speaking Stick** see-and-speak layer are on the roadmap, not in this build.
- Health data stays on the device and laptop; nothing is sent to the cloud.

In one line: Moon Walk — WHOOP for your walking aid: clip it onto the cane or walker you already use, and it tracks your normal, tells you when your walking changes, and coaches your rhythm in real time — all without sending your health to the cloud.

‡ MEDICAL CLAIM SAFETY — Moon Walk is a wellness awareness cue, not a medical assessment.